



Introduction to Deep Equilibrium Nets (DEQNs)

Mini-Course · March 2026

Galo Nuño (Banco de España) · Simon Scheidegger (HEC, University of Lausanne)

FOUNDATIONS & METHODOLOGY

Part I — ML / DL Foundations

DNNs as Function Approximators
Activation Functions & Layers
Stochastic Gradient Descent
Loss Functions & Training

Slides: Part I

Part II — Deep Equilibrium Nets

Motivation & Curse of Dimensionality
DEQN Loss Function Design
Training Algorithm
Supervised vs. Unsupervised Cost

Slides: Part II

Part III — Brock-Mirman Benchmark

Analytical Solution for Validation
DEQN Implementation
Stochastic Extensions (TFP)
Parallelization & Scalability

Notebooks 01 – 03

TOOLS & SCALING

Part IV — NAS & ReLoBRaLo

Neural Architecture Search
Hyperparameter Tuning
Multi-Task Loss Balancing
ReLoBRaLo Normalization

Notebook 05

Part V — Scaling Up: IRBC Model

Multi-Country Dynamics
Equilibrium System & Complementarity
Gauss-Hermite Quadrature
High-Dimensional DEQNs

Notebook 04

Part VI — Deep Surrogate Models

Pseudo-States & Speed Gains
Structural Estimation & UQ
Gaussian Processes & BAL
Implied Volatility Surface

Notebooks 06 – 07

Part VII — Hands-On Exercises & Suggested Readings

Coding exercises (fill-in) · Model implementation · Individual discussions · Notebooks 01 – 07

■ Theory ■ Economic Methods ■ Computational Tools ■ Practice

Tools:

Python

TensorFlow

TF Probability

PyTorch

NumPy

SciPy

Matplotlib

scikit-learn